

A Short Guide to Lemma D.4

Tensorizing the one-mode conic stochastic bound

Abstract

Lemma D.4 lifts the one-mode stochastic inequality of Lemma D.3 to a multimode map, even when that map couples the modes. It does not assert a full multivariate stochastic order. It proves exactly what is needed later: an inequality for products of decreasing photon-number weights. The proof turns all but one mode into a weighted environment, applies Lemma D.3, and then repeats the construction mode by mode.

1 The issue addressed by the lemma

Let

$$\Phi_0 = \tau_{q_1} \otimes \cdots \otimes \tau_{q_r}$$

be a product of thermal states. A multimode channel Γ need not be a product channel: it may create correlations among the output modes. Hence a one-mode result cannot be tensorized merely by multiplying r inequalities.

Lemma D.4 overcomes this by inserting the weights on the untested modes into a new one-mode completely positive map. The trace of that map records the weighted expectation still left to control. Applying the one-mode result successively strips off one factor at a time.

2 Background and notation

2.1 Thermal modes and their amplified reference states

For the number operator $\hat{N}$ on one-mode Fock space $\mathcal{F}$, let

$$\tau_q = (1 - q) \sum_{k \geq 0} q^k |k\rangle\langle k|, \quad 0 < q < 1.$$

Fix an intensity gain $\gamma > 1$ and write

$$q^{(\gamma)} = 1 - \frac{1 - q}{\gamma}.$$

For r modes, define

$$\Phi_0 = \bigotimes_{\ell=1}^r \tau_{q_\ell}, \quad \tau^{(\gamma)} = \bigotimes_{\ell=1}^r \tau_{q_\ell^{(\gamma)}}. \tag{1}$$

The joint number basis is indexed by $\mathbf{k} = (k_1, \dots, k_r)$.

2.2 Separate covariance in every mode

Write

$$D(z) = \bigotimes_{\ell=1}^r D(z_\ell), \quad z \in \mathbb{C}^r.$$

The multimode map Γ is displacement covariant with amplitude gain $\sqrt{\gamma}$ when

$$\Gamma(D(z)XD(z)^*) = D(\sqrt{\gamma}z)\Gamma(X)D(\sqrt{\gamma}z)^*. \quad (2)$$

It is phase covariant in each mode when it intertwines all independent rotations

$$\bigotimes_{\ell=1}^r e^{i\theta_\ell \hat{N}_\ell}.$$

Setting all coordinates of z except z_j equal to zero shows that (2) contains a one-mode covariance relation for each mode. This observation is what makes the reduction below work.

2.3 Product test functions

The lemma considers

$$W(\hat{\mathbf{N}}) = \prod_{\ell=1}^r w_\ell(\hat{N}_\ell), \quad (3)$$

where each $w_\ell : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}_+$ is bounded and decreasing. Because the factors act on different modes, they commute and the product is a bounded positive operator.

This restricted class is important. Ordinary one-dimensional stochastic orders do not automatically imply an order for every decreasing function of the vector $(N_1, \dots, N_r)$ when the output modes may be correlated. The factorization in (3) is precisely what permits a one-mode-at-a-time proof.

3 The one-mode input: Lemma D.3

If Λ is a one-mode, completely positive trace-nonincreasing map with the same displacement and phase covariance, Lemma D.3 states that

$$\mathrm{Tr}[\Lambda(\tau_q)w(\hat{N})] \leq \mathrm{Tr} \Lambda(\tau_q) \mathrm{Tr}[\tau_{q^{(\gamma)}}w(\hat{N})] \quad (4)$$

for every bounded nonnegative decreasing w . That lemma uses the external Guță–Matsumoto least-noise stochastic-order theorem, together with the internally proved constant-trace Lemma D.1. Lemma D.4 introduces no new external input.

4 Statement of Lemma D.4

Lemma D.4 (rephrased). *Let Γ be an r -mode completely positive trace-nonincreasing map, displacement covariant with gain $\sqrt{\gamma}$, and phase covariant in every mode. For the states in (1) and every product weight (3),*

$$\mathrm{Tr}[\Gamma(\Phi_0)W(\hat{\mathbf{N}})] \leq \mathrm{Tr} \Gamma(\Phi_0) \mathrm{Tr}[\tau^{(\gamma)}W(\hat{\mathbf{N}})]. \quad (5)$$

Since both the state and the weight on the right are products,

$$\mathrm{Tr}[\tau^{(\gamma)}W] = \prod_{\ell=1}^r \mathrm{Tr}[\tau_{q_\ell^{(\gamma)}}w_\ell(\hat{N}_\ell)]. \quad (6)$$

5 Proof walkthrough

Step 1: normalize the weights

By multiplying back their suprema after the proof, it is enough to treat $0 \leq w_\ell \leq 1$. If one of the weights is identically zero, the claim is immediate. Otherwise replace w_ℓ by $w_\ell/\|w_\ell\|_\infty$; both sides of (5) acquire the same factor $\prod_\ell \|w_\ell\|_\infty$.

The upper bound by one ensures that every product of the weights is an effect. Inserting its square root therefore preserves the trace-nonincreasing property.

Step 2: package the remaining modes into a one-mode map

The manuscript first treats mode 1. Put

$$W_{>1} = \prod_{\ell=2}^r w_\ell(\hat{N}_\ell), \quad \Phi_{0,>1} = \bigotimes_{\ell=2}^r \tau_{q_\ell},$$

and define

$$\Gamma_1(X) = \text{Tr}_{2,\dots,r} \left[(I \otimes W_{>1}^{1/2}) \Gamma(X \otimes \Phi_{0,>1}) (I \otimes W_{>1}^{1/2}) \right]. \quad (7)$$

This is a one-mode map: prepare the other input modes thermally, apply Γ , test the other output modes with $W_{>1}$, and discard them.

Every operation in (7) is completely positive. For $X \geq 0$,

$$\begin{aligned} \text{Tr} \Gamma_1(X) &= \text{Tr}[\Gamma(X \otimes \Phi_{0,>1})(I \otimes W_{>1})] \\ &\leq \text{Tr} \Gamma(X \otimes \Phi_{0,>1}) \leq \text{Tr} X. \end{aligned}$$

Thus Γ_1 is trace nonincreasing.

Step 3: verify that the induced map has the right symmetries

Apply the multimode covariance relation with displacement vector $(z, 0, \dots, 0)$. The resulting output displacement acts only on mode 1, so it commutes with $I \otimes W_{>1}^{1/2}$ and passes through the partial trace. Consequently,

$$\Gamma_1(D(z)XD(z)^*) = D(\sqrt{\gamma}z)\Gamma_1(X)D(\sqrt{\gamma}z)^*.$$

The same calculation with a phase rotation in mode 1 proves phase covariance. Therefore Γ_1 satisfies every hypothesis of Lemma D.3.

Step 4: remove the first weight

The two traces associated with (7) are

$$\text{Tr}[\Gamma_1(\tau_{q_1})w_1(\hat{N}_1)] = \text{Tr}[\Gamma(\Phi_0)W(\hat{\mathbf{N}})], \quad (8)$$

$$\text{Tr} \Gamma_1(\tau_{q_1}) = \text{Tr}[\Gamma(\Phi_0)W_{>1}]. \quad (9)$$

Applying (4) and substituting these identities gives

$$\text{Tr}[\Gamma(\Phi_0)W] \leq \text{Tr}[\tau_{q_1(\gamma)}w_1(\hat{N}_1)] \text{Tr}[\Gamma(\Phi_0)W_{>1}]. \quad (10)$$

Step 5: iterate

For complete precision, define

$$W_{\geq j} = \prod_{\ell=j}^r w_\ell(\hat{N}_\ell), \quad W_{>j} := W_{\geq j+1}, \quad W_{\geq r+1} = I,$$

with identity factors on the modes not displayed. The one-mode map used at stage j is explicitly

$$\Gamma_j(X) = \text{Tr}_{\ell \neq j} \left[W_{>j}^{1/2} \Gamma \left(\bigotimes_{\ell < j} \tau_{q_\ell} \otimes X \otimes \bigotimes_{\ell > j} \tau_{q_\ell} \right) W_{>j}^{1/2} \right].$$

Here the partial trace discards every output mode except j . The same complete-positivity, trace, and covariance checks used for Γ_1 apply. The weighted and unweighted traces in Lemma D.3 are respectively the two terms in

$$\text{Tr}[\Gamma(\Phi_0)W_{\geq j}] \leq a_j \text{Tr}[\Gamma(\Phi_0)W_{\geq j+1}], \quad a_j := \text{Tr}[\tau_{q_j^{(\gamma)}} w_j(\hat{N}_j)]. \quad (11)$$

Iterating (11) yields

$$\text{Tr}[\Gamma(\Phi_0)W] \leq \left(\prod_{j=1}^r a_j \right) \text{Tr} \Gamma(\Phi_0).$$

Equation (6) identifies the product with $\text{Tr}[\tau^{(\gamma)}W]$, proving (5).

6 A two-mode version

For $r = 2$, the proof is especially transparent. It first gives

$$\text{Tr}[\Gamma(\tau_{q_1} \otimes \tau_{q_2})(w_1 \otimes w_2)] \leq a_1 \text{Tr}[\Gamma(\tau_{q_1} \otimes \tau_{q_2})(I \otimes w_2)].$$

Applying the same construction to mode 2 gives

$$\text{Tr}[\Gamma(\tau_{q_1} \otimes \tau_{q_2})(I \otimes w_2)] \leq a_2 \text{Tr} \Gamma(\tau_{q_1} \otimes \tau_{q_2}).$$

The output may be correlated throughout; the proof never assumes that Γ factors. It uses only the product form of the input and test weight, plus separate covariance in each mode.

7 What the lemma does not claim

Lemma D.4 proves a conic inequality: the factor $\text{Tr} \Gamma(\Phi_0) \leq 1$ keeps track of subnormalization. It does not claim that $\Gamma(\Phi_0)$ is below $\tau^{(\gamma)}$ in operator order, nor that its full joint photon-number law stochastically dominates the product geometric law for every multivariate decreasing test. The product class in (3) is sufficient because the fidelity witness in Lemma D.5 factorizes mode by mode.

8 Dependencies and role in the paper

The dependency chain is

$$\text{Lemma D.1} + \text{external least-noise theorem} \longrightarrow \text{Lemma D.3} \longrightarrow \text{Lemma D.4}.$$

Beyond D.3, the proof is internal: it constructs the induced map, checks its covariance, and iterates. D.4 supplies the factorized fidelity witness in D.5, hence Theorem 3.4, and is also applied branchwise in the hybrid converse. Its prerequisites are completely positive maps, partial trace, product thermal states, covariance, and Lemma D.3.